# Referee package – STEP-5B rectangle-constant closure: the theorem-grade rectangle prefactor

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

first issued 2026-06-10 · this version issued 2026-06-12 · v1.2

## 1. Purpose and scope

This referee document concerns the STEP-5B rectangle-constant closure: the theorem-grade prefactor $K(n)$ controlling the layer budget. **Scope:** the rectangle constant and the official threshold. **Not claimed:** the alternative 20/9 incidence route is provisional (constant unpinned) and is not relied upon. This package does not by itself close full STEP-5B (the budget comparison and admissibility gates are tracked separately), does not close admissible-class exhaustiveness, and does not enact the full Reading-H $C_{\text{full}}$ theorem.

## 2. Definitions and the result

**Definitions (patch v1.1-1).** Let $Q \subset S^2_{q_0}$ be an admissible pattern of $n$ independent modes on the condensation shell, with symmetrised (antipodal) point set $\hat{Q} = Q \cup (-Q)$ of $N = 2n$ points and amplitudes $\{A_a\}$ of total intensity $I = \sum_a A_a^2$. The rectangle prefactor is the normalised off-diagonal transfer $\ell^2$ mass

$$K(n) \;=\; \frac{1}{(\lambda' I)^2} \sum_{t \neq 0} w_t^2, \qquad w_t \;=\; \lambda' \sum_{\substack{a+b=t \\ a,b \in \hat{Q}}} A_a A_b,$$

i.e. the quantity the STEP-5B layer budget must dominate. The balance parameter $\kappa$ is the registered **H-KBAL** amplitude-balance constant,

$$\textbf{H-KBAL}: \quad A^2_{\max} \;\leq\; \kappa^2\, I/n,$$

with $\kappa \simeq 1$ at the production operating point (equal-amplitude patterns have $\kappa = 1$ exactly).

**The result.** (i) THEOREM (operator verdicts #9/#10): for every $\kappa$-balanced admissible pattern,

$$K(n) \;\leq\; 8 + 4\sqrt{14}\,\kappa^4 \sqrt{n}, \qquad c_R^{\text{rigorous}} = 4\sqrt{14} \simeq 14.97,$$

with $1.59 \times 10^5$ the OFFICIAL threshold (derivation in Sec. 3). (ii) The 20/9 incidence route ($2.2 \times 10^{10}$) is provisional with an unpinned constant and is recorded but not load-bearing.

**Provenance (patch v1.1-3).** The rectangle theorem is imported from operator verdicts #9/#10, archived in the folder note `rectangle-constant-closure-260605-260605-v1.3.tex.txt` §2–§3: verdict #9 supplied the derivation $\sum_C p_C^3 \leq 7N^3$ (carrier-circle cube bound) with the Cauchy–Schwarz interpolation $\sum_C p_C^2 \leq (\sum_C p_C)^{1/2} (\sum_C p_C^3)^{1/2} \leq \sqrt{7}\, N^{5/2}$; verdict #10 repaired the provisional incidence exponent to 20/9 (correcting the 28/13 arithmetic slip). This package records and machine-reproduces the numerical threshold but does not reprove the incidence alternative.

## 3. Structure of the argument and threshold derivation

The rectangle prefactor is bounded by the closed-form $8 + 4\sqrt{14}\,\kappa^4\sqrt{n}$ (the operator-verified rectangle theorem). STEP-5B closes for a pattern of $n$ modes provided (patch v1.1-2)

$$K(n) \;\le\; 8 + 4\sqrt{14}\,\kappa^4\sqrt{n} \;<\; K_{\text{budget}}(I) \;=\; \frac{4\,(1-a_0(I))\,\Delta F_{\text{margin}}}{(\lambda' I)^2 J_{\max}},$$

which at the production anchor $I = 4\times10^{-4}$ (where $K_{\text{budget}} = 5972$, $\kappa \simeq 1$) gives the official admissible-mode threshold

$$n_{\text{adm}} \;\lesssim\; \left(\tfrac{K_{\text{budget}}-8}{4\sqrt{14}\,\kappa^4}\right)^2 \;=\; \left(\tfrac{5972-8}{4\sqrt{14}}\right)^2 \;=\; 1.59\times10^5.$$

($3.77\times10^3$ at $I = 10^{-3}$; $2.02\times10^2$ at $I = 2\times10^{-3}$.) The incidence-geometry alternative would relax this to $2.2\times10^{10}$ but its constant is not pinned, so it is documented for cross-check only and is not used in the closure.

## 4. Numerical reproduction

The beyond-layer Gershgorin bound script reproduces the rectangle constant and the $1.59\times10^5$ threshold with asserts; run it from the bundle. Operator confirmation run (2026-06-12, independent clean-repo reconstruction with the legacy canonical sources `Math424_AddA_reading_uniqueness.py` / `Math400_-AddE_brazovskii_one_loop.py` resolved): 192/192 asserts PASS, exit 0, $c_R = 4\sqrt{14} = 14.9666$, $n_{\text{adm}} = 1.59\times10^5/3.77\times10^3/2.02\times10^2$ at $I = 4\times10^{-4}/10^{-3}/2\times10^{-3}$. AI-side clean-run record (2026-06-12, inspection re-run): 192/192 asserts PASS, exit 0; key asserts `operator_p3_bound` (3 configs), `operator_interpolation` (3 configs), `operator_cR_rigorous` ($c_R = 14.9666$), `operator_-theorem_region_anchor` ($n \lesssim 1.59\times10^5$ at $I = 4\times10^{-4}$), `incidence_exponent_repaired_20_-9`, `incidence_route_region_repaired` (provisional, not load-bearing); JSON artefact refreshed at `claims/B5-BEYOND-LAYER-BOUND/runs/260605-gershgorin-reduction/result.json`.

## 5. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

($\alpha$) *Attack the scope.* The result is confined to the scope stated in Sec. 1; a referee may push that boundary, which is fair — the package makes no claim outside it, and the precise scope is in the footer. In particular the package does NOT claim full STEP-5B closure, admissible-class exhaustiveness, or the full $C_{\text{full}}$ theorem.

($\beta$) *Attack the tier / over-claim.* No prose here is stronger than the registered grade (footer no-overclaim line); the status is PUBLISHED-BUNDLE CONFIRMED within the rectangle-prefactor scope (T6 support package, NOT full STEP-5B closure and NOT T7), and the claim-ledger tier is the arbiter.

($\gamma$) *Attack reproduction / self-containment.* Each reproduction script carries self-test asserts and ships in the folder's bundle with its transitive dependencies (imports and runtime file reads, resolver v1.3.0); a referee runs them and diffs against `expected/`.

Quantitative sanity check (mandatory): the threshold is reproduced from its inputs, $((5972-8)/(4\sqrt{14}))^2 = 1.588\times10^5 \approx 1.59\times10^5$ (rounding consistent with the operator's independent $1.58\times10^5$); the falsifier (footer) is concrete and pre-registered, so the claim is refutable by a single explicit construction; and the numerical values cited are reproduced by the folder's script (192/192 asserts pass, exit 0, 2026-06-12 re-run).

## 6. Result footer

| | |
|---|---|
| Result ID: | B5-BEYOND-LAYER-BOUND / STEP-5B rectangle-constant referee package |
| Precise statement: | THEOREM (operator verdicts #9/#10, provenance: rectangle-constant-closure v1.3 S2-S3): for kappa-balanced patterns (H-KBAL: A_max^2 <= kappa^2 I/n), K(n) = sum_{t!=0} w_t^2/(lam' I)^2 <= 8 + 4 sqrt(14) kappa^4 sqrt(n); official threshold n_adm <~ ((K_budget-8)/(4 sqrt 14))^2 = ((5972-8)/(4 sqrt 14))^2 = 1.59e5 at the anchor I=4e-4. The 20/9 incidence route (2.2e10) is provisional (constant unpinned), not load-bearing. |
| Scope: | the STEP-5B rectangle prefactor K(n) and official threshold. Does NOT close full STEP-5B, admissible-class exhaustiveness, or full C_full. |
| Evidence grade: | theorem-grade rectangle constant (T6 support package); EXECUTED (192/192 asserts, exit 0; operator independent clean-run CONFIRMED). |
| Reproduction command: | python codes/vacuum/beyond_layer_gershgorin_bound.py |
| Falsifier: | a kappa-balanced configuration where K(n) exceeds 8 + 4 sqrt(14) kappa^4 sqrt(n); or the layer budget failing the 1.59e5 threshold. |
| Tier before / after: | PUBLISHED-BUNDLE CONFIRMED (operator CONFIRM 2026-06-12); bundle STEP-5B-Rectangle-T6-260612; T6 support package (not T7: no independent full STEP-5B closure). No B5 claim-tier promotion. |
| No-overclaim statement: | the rectangle constant is theorem-grade within the H-KBAL scope; the 20/9 incidence route is provisional and not relied upon; full STEP-5B closure is not claimed. |
| Next required action: | none for this package (PUBLISHED). Mainline: SC-SCOPE and DR-2 referee packages; full STEP-5B budget comparison + admissibility gates remain separately tracked. |